

**Real-World Order of Operations**

Question Bank - Additional 4.07

Name \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

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*Show your work. Circle your final answer.*

**For questions 1 through 4, substitute and evaluate.**

1.  $5x^2$  when  $x = -4$

2.  $\frac{1}{2}mv^2$  when  $m = 30$  and  $v = 10$

3.  $\frac{a^2+b}{c}$  when  $a = 8$  ,  $b = 4$  ,  $c = 17$

4.  $3(x+y)^2$  when  $x = -5$  and  $y = 2$

**For questions 5 and 6, write and evaluate an expression. Give the unit.**

5. A square room of side 13 feet is floored at 2.75 dollars a square foot, with a flat 90 dollars for delivery. Find the total cost.

6. A rectangular patio 16 feet by 7 feet is sealed at 1.40 dollars a square foot, with a flat 35 dollars for equipment hire. Find the total cost.

**Two formulas.**

7. Body mass index is  $B = \frac{703w}{h^2}$ , where  $w$  is a weight in pounds and  $h$  is a height in inches. Find  $B$  to the nearest tenth.

a.  $w = 120$  ,  $h = 62$

b.  $w = 165$  ,  $h = 68$

c.  $w = 210$  ,  $h = 74$

8. Kinetic energy is  $K = \frac{1}{2}mv^2$ , for mass  $m$  in kilograms and speed  $v$  in metres per second. Find  $K$  in joules.

a.  $m = 1,500$ ,  $v = 20$

b.  $m = 640$ ,  $v = 9$

c.  $m = 55$ ,  $v = 4$

**Working backwards.**

9. Find the missing value in each situation.

a.  $\frac{1}{2}mv^2 = 1,600$  and  $v = 8$ . Find  $m$ .

b.  $x^2 - 7 = 74$ , where  $x$  is positive. Find  $x$ .

- c. A square lawn of side  $n$  feet is turfed at 3.00 dollars a square foot with a flat 55 dollars delivery, and the total is 1,255 dollars. Find  $n$ .

**Reading the situation.**

10. Two problems use the numbers 8, 12 and 4. **A:** a printer costs 8 dollars to hire plus 12 dollars for each of 4 days. **B:** a printer costs 8 dollars for each of 12 reams, plus 4 dollars for delivery. Find both totals, and explain why circling the numbers would not have told you what to do.
11. A square court of side 20 feet is resurfaced at 3.60 dollars a square foot, with a flat 200 dollars for the equipment. Which expression gives the total cost?
- ☐ A.  $20^2(3.60) + 200$
  - ☐ B.  $20^2(3.60 + 200)$
  - ☐ C.  $20(3.60) + 200$
  - ☐ D.  $20^2 + 3.60 + 200$
12. Select ALL of the expressions whose value is 32 when  $x = 4$ .
- ☐ A.  $2x^2$
  - ☐ B.  $(2x)^2$
  - ☐ C.  $x^2 + 16$
  - ☐ D.  $\frac{128}{x}$
  - ☐ E.  $x^2 \cdot 2$
  - ☐ F.  $2x \cdot 2$